

How to create a Flexible Fastener on 24X

- [Points creation](#)
- [Fasteners Creation](#)

To create a Flexible Fastener, the following structure must be created:

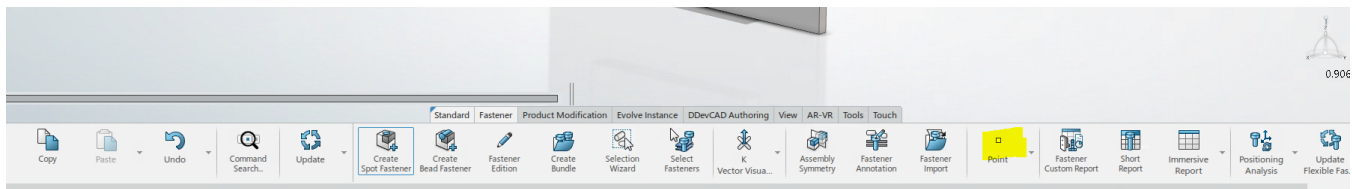
- EM above an SA, above an EP: [How to create upper and lower level entities](#)
- The 3D shape has also be created underneath the EP: [How to create a 3D shape](#)

Once this hierarchy is done, we need to create "Point" function: These points are used to identify the possible positions for the fasteners.

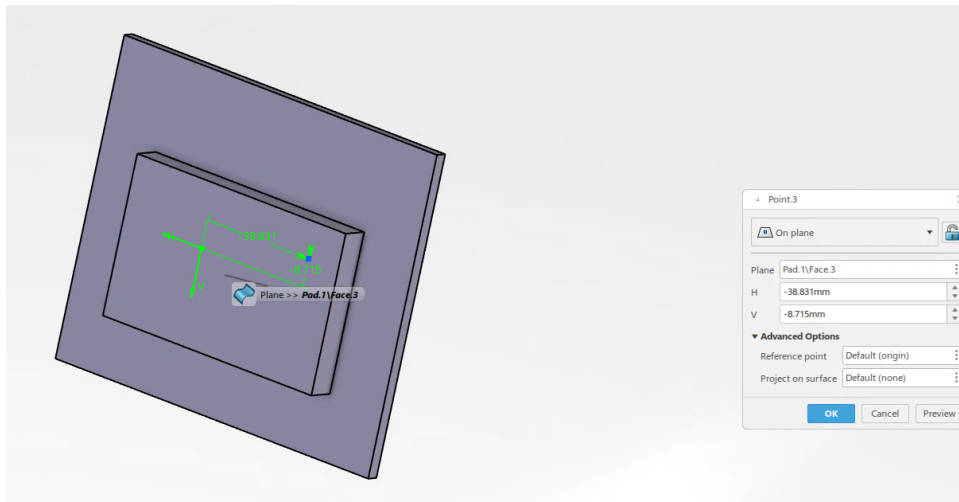
To create "Point" function:

Points creation

- Double click on one of 3D Shape previously created
- Click on "Point" function in the toolbar



- Click on OK
- Select a surface of the pad



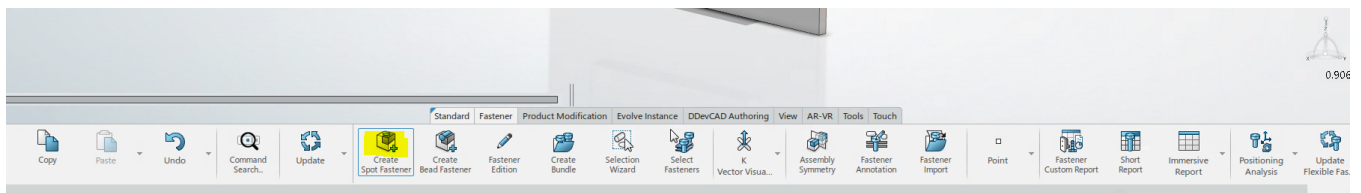
- Click on OK
- Repeat the above actions to create other points

In our case, we created 3 points.

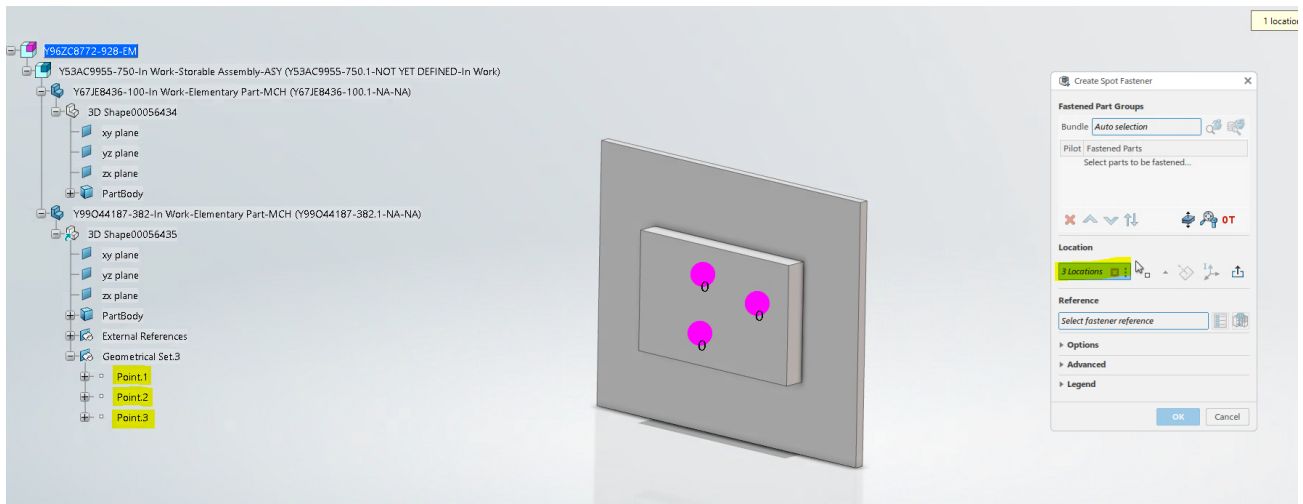
Fasteners Creation

In the Compass on top left

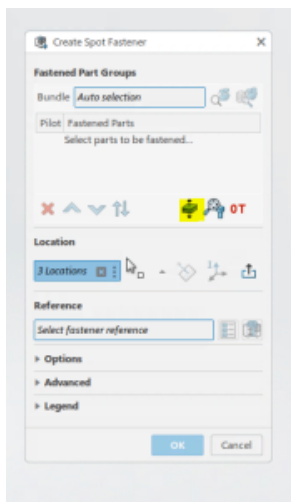
- Launch the application "Fastening"
- Select the main node of the assembly : it could be an EM or an Storable Assembly
- Click on "Create Spot Fastener" function in the toolbar



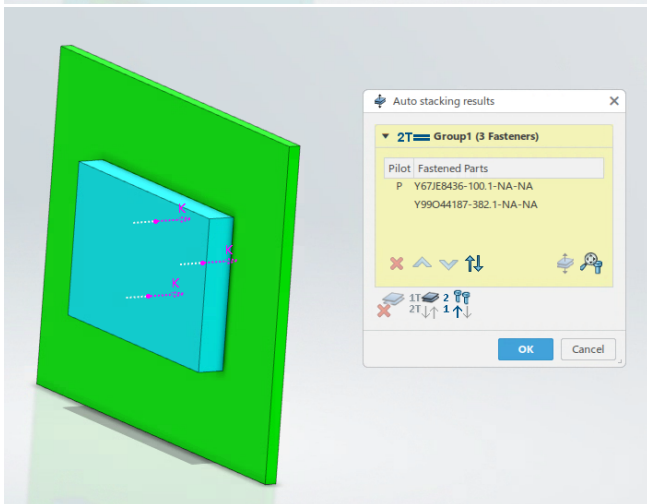
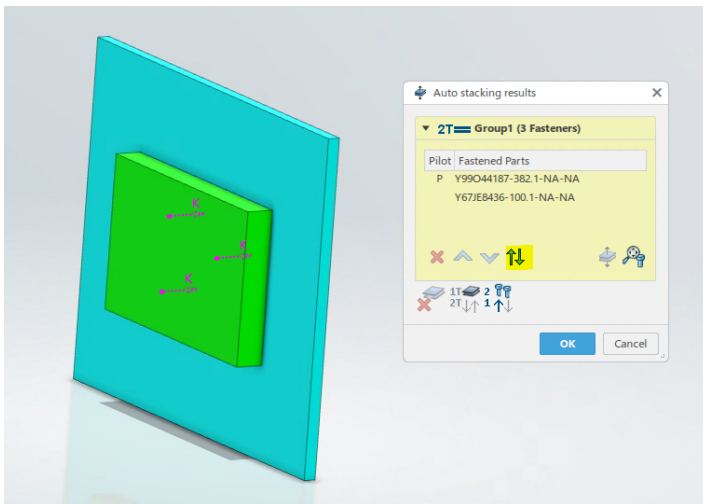
- For the location Select 3 points for instantiation



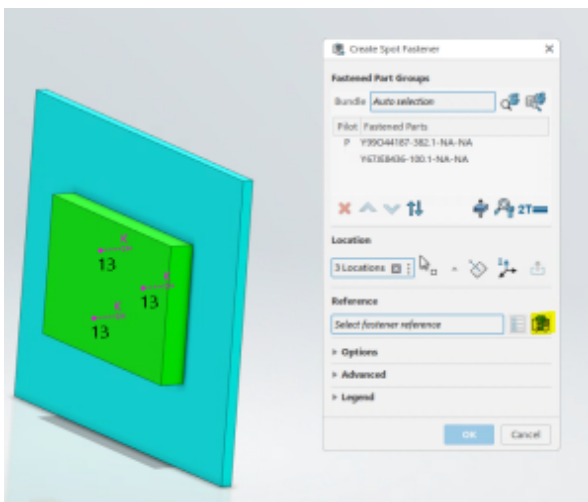
- Click on "Automatic Stacking" function



- We can click on these two arrows to reverse stack part and click OK



- Click on "Fastener Composition Assistant Tool"



- Select for example: Any material

Fastener attribute and Components selection

Specification: Eurodrone_Fastening_for_Airbus_Defence_and_Space

Stacking material: ☐ Automatic Not found ☒ Manual **Select**

Process category: Not found

Process type: Not found

Diameter: Not found

Custom Attributes

Fastener Head Type: No results

Fastener Shank Material: No results

Fastener Standard Family: No results

Material code and Finish code: No results

Components

Configuration: Not found

OK Cancel

To fill these informations, we need to refer to a list of classified items:

to acces to this list:

- Open "Classify and Reuse" Application on the compass
- Click on the library" Screws and bolts"

3DEXPERIENCE | ENOVIA Classify and Reuse

Search

Libraries: Filter: All

Title	Purpose	Rev	Type	Name	Description	State
Anchor Nuts	Definition	-	General Library	Anchor Nuts		Active
ATA-ASD Code	Definition	-	General Library	ATA-ASD Code		Active
Common Ergo Posture Library	Definition	-	General Library	Common Ergo Posture ...		Active
Control Plan Generative Instructions	Definition	-	General Library	Control Plan Generativ ...		Active
CONTROLLED US EXTRA-TERRITORIAL EXPORT CONTROL		-	Export Control Library	CONTROLLED US EX ...		Active
Drill / PreDrill	Definition	-	General Library	Library-001FFA11-0000 ...	Library to test the modifi...	Active
EU Dual Use	Definition	-	General Library	EU Dual Use		Active
Export Control Classification Airbus Commercial	Definition	-	General Library	Export Control Classific...		Active
NATIONAL AND EU EXPORT CONTROL	Definition	-	General Library	NATIONAL AND EU EX...		Active
NATIONAL AND EU EXPORT CONTROL	Definition	-	General Library	NATIONAL AND EU EX...		Active
Nuts and Collars	Definition	-	General Library	Nuts and Collars		Active
OCCAR	Definition	-	General Library	OCCAR		Active
Quality Authorization Library MAN	Definition	-	General Library	Quality Authorization Li...		Active
Quality Authorization Library TAB	Definition	-	General Library	Quality Authorization Li...		Active
Rivets	Definition	-	General Library	Rivets		Active
Screws and Bolts	Definition	-	General Library	Screws and Bolts		Active
SECURITY CLASSIFICATIONS	Definition	-	General Library	SECURITY CLASSIFIC...		Active
SECURITY CLASSIFICATIONS	Definition	-	General Library	SECURITY CLASSIFIC...		Active
Standard Parts Non Aero	Definition	-	General Library	Standard Parts Non Aero		Active
TEST		-	Export Control Library	ECL-000000004		Active
test FACS library	Definition	-	General Library	Library-001FFA11-0000 ...		Obsolete

- Click on this class "Screws and Bolts"

3DEXPERIENCE | ENOVIA Classify and Reuse

Search

Screws and Bolts

General Library (-)

Maturity State: In Work + **Active** + Obsolete

Owner: Admin Platform

Modified: 2025 May 23 16:54:35

Drop files here

Screws and Bolts(101)

Sharing

Discussions

Classes

History

Lifecycle

Routes

Referenced By

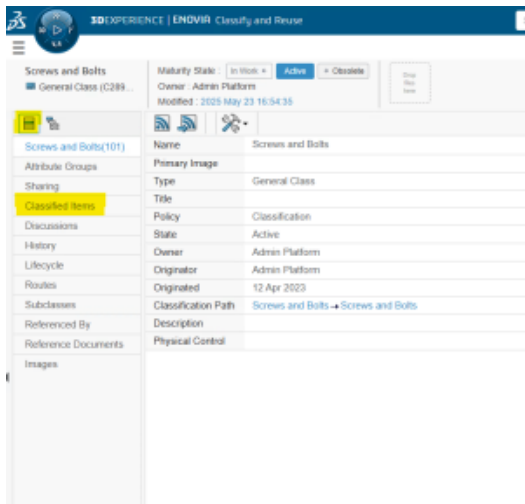
Reference Documents

Background Jobs

Images

Title	Type	Name
Screws and Bolts	General Class	Screws and Bolts

- On the category on the left, click on Classified Items to open the list



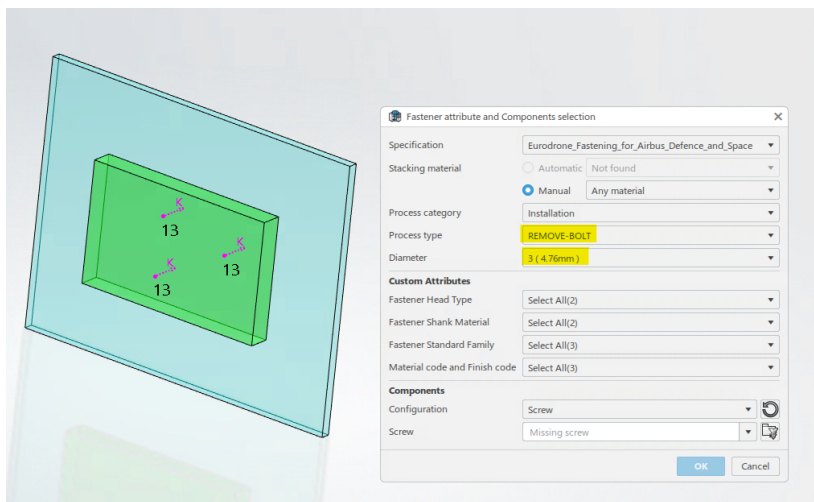
- From this list, we can fill these informations:
 - Process type
 - Diameter with the "Diameter Code" and "Nominal diameter"

Title	Grip length min	Grip length max	Process type	Diameter code	Nominal diameter	Name
Doc-0000005	0.0 m	0.0 m			0.0 m	Doc-0000005
Drawing0000442	0.0 m	0.0 m			0.0 m	Drawing-001FFA11-0...
	0.0 m	0.0 m			0.0 m	Doc-0000009
	0.0 m	0.0 m			0.0 m	Milestone1
ABS0114T3-2	1.5875 mm	3.175 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-3	3.175 mm	4.7625 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-4	4.7625 mm	6.35 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-5	0.0 m	0.0 m			0.0 m	Doc-001FFA11-00...
ABS0114T3-6	7.9375 mm	9.525 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-7	9.525 mm	11.1125 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-8	11.1125 mm	12.7 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-9	12.7 mm	14.2875 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-10	14.2875 mm	15.875 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-11	15.875 mm	17.4625 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-12	17.4625 mm	19.05 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-13	19.05 mm	20.6375 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-14	20.6375 mm	22.225 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...
ABS0114T3-15	22.225 mm	23.8125 mm	REMOVE-BOLT	3	4.76 mm	Doc-001FFA11-00...

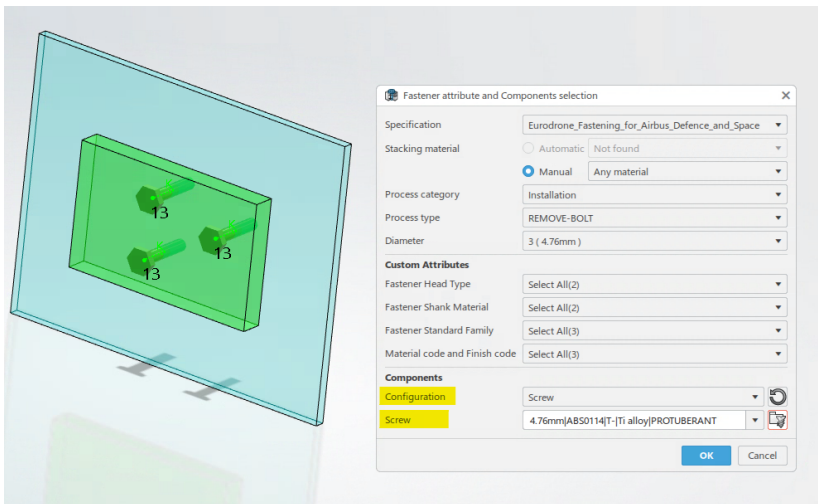
These informations are based on the thickness of the part, so you need to see the interval between Grip length min and Grip length max:

For our example:

- Process Type: REMOVE-BOLT
- Thickness = 13 mm > between 12.7 mm and 14.2875 mm
- Diameter: 3 (4,76mm)



- So we choose our Components:
 - For Configuration: Screw,
 - For Screw: Select any from drop down list to choose the type of screw



Click on OK

the fasteners are created

